MONOMORPHISMS APPEAR AT A FINITE STAGE IN THE DESCENT SITUATION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1955-1966, following the statement at 1945-1953.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Diagonal is locally of finite presentation. Morphisms of Spaces, Lemma spaces-morphisms-lemma-diagonal-morphism-finite-type (cross-chapter, cited at ~1955-1958). X_0 -> X_0 x_{Y_0} X_0 is lfp.
- Isomorphisms appear at a finite stage. lemma-descend-isomorphism (same chapter, cited at ~1960-1963). Applied to the diagonal: Delta_i is an isomorphism for some i.

Source reference. Original lemma 0852: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1925–1925. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084W: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1666–1666. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 084P: focused-seeds/geometry/S9-spaces-limits-descend-monomorphism/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Notation and assumptions as in Situation 084W. If*

(1) *f is a monomorphism,*
(2) *f_0 is locally of finite type,*

then f_i is a monomorphism for some $i \geq 0$.

Proof. Recall that a morphism is a monomorphism if and only if the diagonal is an isomorphism. The morphism $X_0 \to X_0 \times_{Y_0} X_0$ is locally of finite presentation by Morphisms of Spaces, Lemma 084P. Since $X_0 \times_{Y_0} X_0$ is quasi-compact and quasi-separated we conclude from Lemma 0852 that $\Delta_i : X_i \to X_i \times_{Y_i} X_i$ is an isomorphism for some $i \geq 0$. For this i the morphism f_i is a monomorphism. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.